

IWT Assignment Answers

Answer 1:

IoT Architecture consists of sensors, devices, gateways, cloud and applications. Sensors collect data from environment. Gateway transfers data to cloud using internet. Cloud stores and processes data. User can monitor devices using mobile application. This architecture helps in automation and smart decision making.

Answer 2:

Sensors are devices that detect physical changes like temperature, humidity and motion. Actuators perform actions based on received signals. Example: motor, relay and buzzer. Sensors collect input while actuators provide output in IoT systems.

Answer 3:

Cloud Computing provides storage, processing and remote access in IoT. IoT devices send data to cloud servers where analysis and monitoring is done. Cloud improves scalability, security and real time access.

Answer 4:

MQTT stands for Message Queuing Telemetry Transport. It is a lightweight communication protocol used in IoT. MQTT works on publish and subscribe model and consumes very low bandwidth.

Answer 5:

Smart Home Automation system uses IoT devices to control lights, fans, security and appliances. Users can control devices remotely using mobile applications and internet. It increases comfort, security and energy efficiency.